Yaoyang Liu

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Education

The Hong Kong University of Science and Technology (HKUST) **Hong Kong**
PhD in Computer Science and Engineering *2025–Present*
- Research Focus: Video Generation, Computer Vision
- Advisor: Prof. Long Chen

University of Science and Technology of China (USTC) **Hefei, China**
Bachelor of Engineering in Artificial Intelligence *2021–2025*
- School of Information Science and Technology
- GPA: 4.13/4.3, Rank: 1/60

Research Experience

Advised by Prof. Jiebo Luo and Prof. Li Yuan **University of Rochester (Remote)**
Research on Multi-modal Large Language Models *2024*
- Participated in the ChronoMagic-Bench project as part of open-sora-plan
- Conducted data processing and benchmark development for metamorphic evaluation of text-to-time-lapse video generation
- Studied diffusion models and gained foundational knowledge in multi-modal large language models

Publications

- **Yaoyang Liu***,, Dongyu Yan*, Yihua Du, Shengju Qian, Zeke Xie, Xin Wang, Ying-Cong Chen, Long Chen. "RAR: Rate-Adaptive Rollout for Time-Consistent Interactive World Models" *Preprint*, 2026. (*Equal contribution)
- **Yaoyang Liu**, Yuechen Zhang, Wenbo Li, Yufei Zhao, Rui Liu, Long Chen. "SwiftI2V: Efficient High-Resolution Image-to-Video Generation via Conditional Segment-wise Generation." *Preprint*, 2026.
- Yuechen Zhang*, **Yaoyang Liu***, Bin Xia, Bohao Peng, Zexin Yan, Eric Lo, Jiaya Jia. "Magic Mirror: ID-Preserved Video Generation in Video Diffusion Transformers." *IEEE International Conference on Computer Vision (ICCV)*, 2025. (*Equal contribution)
- Yiming Zhao, Yu Zeng, Yukun Qi, **Yaoyang Liu**, Lin Chen, Zehui Chen, Xikun Bao, Jie Zhao, Feng Zhao. "V2P-Bench: Evaluating Video-Language Understanding with Visual Prompts for Better Human-Model Interaction." *International Conference on Learning Representations (ICLR)*, 2025.
- Shenghai Yuan, Jinfa Huang, Yongqi Xu, **Yaoyang Liu**, Shaofeng Zhang, Yujun Shi, Ruijie Zhu, Xinhua Cheng, Jiebo Luo, Li Yuan. "ChronoMagic-Bench: A Benchmark for Metamorphic Evaluation of Text-to-Time-lapse Video Generation." *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track (Spotlight)*, 2024.

Honors & Awards

USTC Guo Moruo Scholarship	2025
Undergraduate National Scholarship	2023
Artificial Intelligence Talent Class Scholarship	2023
Robogame Robot Competition, First Prize	2023
Undergraduate National Scholarship	2022

Technical Skills

ML/AI Frameworks: PyTorch, Diffusion Models, Transformers
Research Areas: Video Generation, Multi-modal Large Language Models, Computer Vision